

# Project ARGUS — Optimization Results

COTS Enhanced | Recommended Design Configuration

|                            |                         |                        |                                   |                              |                             |
|----------------------------|-------------------------|------------------------|-----------------------------------|------------------------------|-----------------------------|
| <b>\$7,458</b><br>per unit | <b>68</b><br>fleet size | <b>99%</b><br>swarm Pd | <b>\$507K</b><br>total fleet cost | <b>11.5 kg</b><br>total mass | <b>30 days</b><br>endurance |
|----------------------------|-------------------------|------------------------|-----------------------------------|------------------------------|-----------------------------|

## Airframe

| Parameter       | Value              |
|-----------------|--------------------|
| Wingspan        | 8.0 m              |
| Chord           | 0.30 m             |
| Wing area       | 2.4 m <sup>2</sup> |
| Aspect ratio    | 26.7               |
| Structural mass | 1.92 kg            |
| L/D ratio       | 48.9               |
| Cruise speed    | 51.9 kts (96 km/h) |
| Stall speed     | 43.1 kts           |
| Mach            | 0.090              |

## Sensor suite

| Component                  | Cost    | Function            |
|----------------------------|---------|---------------------|
| MEMS infrasound x8         | \$500   | Shockwave detection |
| SWIR InGaAs array          | \$1,500 | Thermal tracking    |
| Jetson Orin Nano           | \$250   | On-board ML         |
| EO camera (stabilized)     | \$300   | Visual verification |
| Multi-band radio + GPS/INS | \$400   | Comms & nav         |

Single-node detection probability: 65% | Detection range: 75 km | Payload mass: 2.8 kg | Power draw: 21 W continuous

## Power system

| Parameter        | Value                          |
|------------------|--------------------------------|
| Solar area       | 1.92 m <sup>2</sup> (GaAs 30%) |
| Peak solar power | 762 W                          |
| Usable avg solar | 399 W                          |

| Parameter                      | Value                    |
|--------------------------------|--------------------------|
| Battery                        | 1,379 Wh (Li-S, 3.96 kg) |
| Cruise power                   | 74.9 W                   |
| Total power (cruise + payload) | 95.9 W                   |
| Power margin                   | 117%                     |

## Fleet & coverage (500,000 km²)

| Parameter                             | Value                        |
|---------------------------------------|------------------------------|
| Active nodes                          | 43                           |
| Total fleet (incl. rotation + spares) | 68                           |
| Coverage per node                     | 17,671 km²                   |
| Overlap                               | 1.5x (avg 2 nodes per point) |
| Swarm fused Pd                        | 99%                          |
| Fleet cost                            | \$507,111                    |
| Cost per km²                          | \$1.01                       |
| Annual maintenance                    | \$50,711                     |
| Airframe lifetime                     | 3.0 years                    |

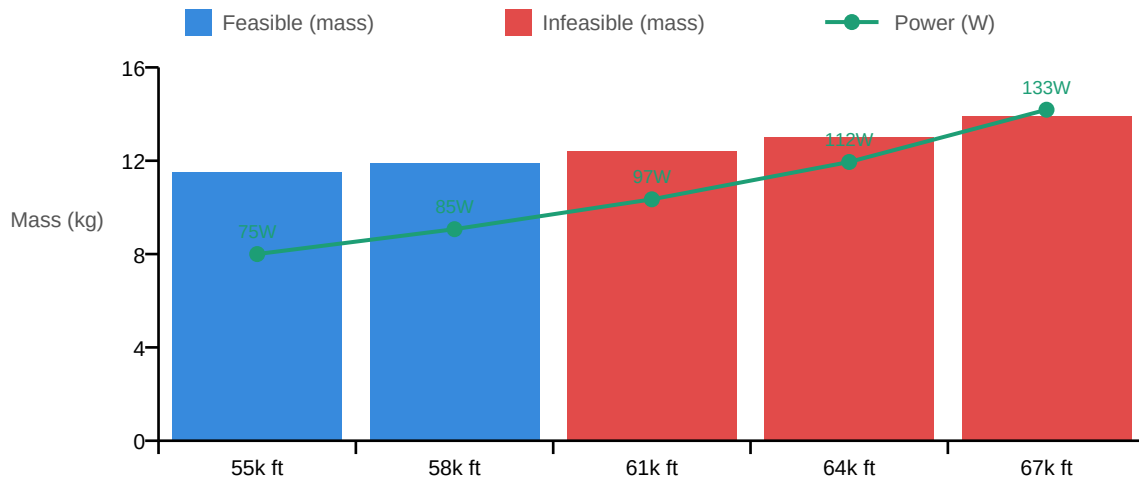
## Per-unit cost breakdown

| Item                            | Cost    | Share |
|---------------------------------|---------|-------|
| Sensor payload                  | \$5,400 | 72.4% |
| Assembly / integration / test   | \$500   | 6.7%  |
| Airframe (CF composite)         | \$384   | 5.1%  |
| Propulsion (2x BLDC)            | \$300   | 4.0%  |
| Battery (Li-S)                  | \$207   | 2.8%  |
| Power electronics               | \$200   | 2.7%  |
| Recovery hardware               | \$200   | 2.7%  |
| Tooling (amortized / 200 units) | \$190   | 2.5%  |
| Solar cells                     | \$77    | 1.0%  |

*Total per unit: \$7,458 | Production run: 200 units*

## Altitude sensitivity

Mass and power required vs. altitude. Design is feasible at 55,000–58,000 ft; above 61,000 ft power diverges and mass convergence fails.



### Feasibility checks (all pass)

| Check           | Criterion                                     | Actual                 | Status |
|-----------------|-----------------------------------------------|------------------------|--------|
| Energy feasible | Day surplus > night need                      | 3,789 Wh surplus       | PASS   |
| Power margin    | > 5%                                          | 117%                   | PASS   |
| Stall margin    | $V_{\text{cruise}} / V_{\text{stall}} > 1.15$ | 1.20                   | PASS   |
| Wing loading    | < 5.0 kg/m <sup>2</sup>                       | 4.79 kg/m <sup>2</sup> | PASS   |
| Total mass      | < 250 kg                                      | 11.5 kg                | PASS   |
| Aspect ratio    | 10 – 40                                       | 26.7                   | PASS   |
| Span structural | ≤ 35 m                                        | 8.0 m                  | PASS   |